

Linear Algebra II (2026 Fall)

Lecture 3.

Studying Contents :

Chapter 5. Diagonalization

- Section 5.1 : Eigenvalues and Eigenvectors

Recall : Let $T : V \rightarrow V$ be a linear transformation of an n -dimensional vector space V .

(1) A nonzero vector $v \in V$ is called an **eigenvector** of T if

there exists a scalar $\lambda \in \mathbb{F}$ such that $T(v) = \lambda v$.

The scalar λ is called the **eigenvalue** corresponding to the eigenvector v .

(2) If there exists an ordered basis $\mathcal{B} = \{v_1, v_2, \dots, v_n\}$ for V such that

$$T(v_i) = \lambda_i v_i \text{ for some } \lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{F},$$

then

$$[T]_{\mathcal{B}} = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix} \text{ is a diagonal matrix.}$$

In this case, we say that T is **diagonalizable**.

(3) Let A be an $n \times n$ square matrix. Then we say that A is **diagonalizable** if

the linear transformation $L_A : \mathbb{F}^n \rightarrow \mathbb{F}^n$ is diagonalizable.

Next Problem : How to find eigenvectors and eigenvalues of a given matrix $A \in M_{n \times n}(\mathbb{F})$?

Theorem 5.2. : Let $A \in M_{n \times n}(\mathbb{F})$ and let $\lambda \in \mathbb{F}$. Then

λ is an eigenvalue of A if and only if $\det(A - \lambda I_n) = 0$.

Proof : λ is an eigenvalue of $A \Leftrightarrow$ there exists a nonzero vector $v \in \mathbb{F}^n$ such that $Av = \lambda v$

$$(Av = \lambda v \Leftrightarrow Av = \lambda I_n v \Leftrightarrow (A - \lambda I_n)v = \mathbb{O})$$

$$\Leftrightarrow N(A - \lambda I_n) \text{ is not the zero space and so } \text{rank}(A - \lambda I_n) < n$$

$$\Leftrightarrow A - \lambda I_n \text{ is not invertible}$$

$$\Leftrightarrow \det(A - \lambda I_n) = 0. \quad \blacksquare$$

Definition : For $A \in M_{n \times n}(\mathbb{F})$,

the polynomial $f(t) := \det(A - tI_n)$ is called the **characteristic polynomial** of A .

Remark : Theorem 5.2 states that

the eigenvalues of A are the zeros of its characteristic polynomial $\det(A - tI_n)$.

Definition : Let $T : V \rightarrow V$ be a linear transformation of an n -dimensional vector space V .

Also let $\mathcal{B} = \{v_1, \dots, v_n\}$ be an ordered basis for V .

We define the **characteristic polynomial** $f(t)$ of T to be

the characteristic polynomial of $A = [T]_{\mathcal{B}}$.

That is, $f(t) := \det(A - tI_n)$.

Remark : At first glance,

the characteristic polynomial of T seems to depend on the choice of basis $\mathcal{B} = \{v_1, \dots, v_n\}$.

To address this, let $\gamma = \{w_1, \dots, w_n\}$ be another ordered basis of V . Then

$$[T]_{\gamma} = Q [T]_{\mathcal{B}} Q^{-1} \text{ where } Q = [Id_V]_{\mathcal{B}}^{\gamma}.$$

Therefore,

$$[T]_{\gamma} - tI_n = Q [T]_{\mathcal{B}} Q^{-1} - tI_n = Q ([T]_{\mathcal{B}} - tI_n) Q^{-1}.$$

Taking determinants, we get

$$\det([T]_{\gamma} - tI_n) = \det Q \times \det([T]_{\mathcal{B}} - tI_n) \times \det Q^{-1} = \det([T]_{\mathcal{B}} - tI_n).$$

Hence, the characteristic polynomial of T is independent of the choice of ordered basis.

Accordingly, we denote the characteristic polynomial of T simply as $\det(T - tI)$.

Example 5. : Consider the linear operator

$$T : P_2(\mathbb{R}) \rightarrow P_2(\mathbb{R}), \quad T(f(x)) = f(x) + (x+1)f'(x).$$

For the standard basis $\mathcal{B} = \{1, x, x^2\}$ for $P_2(\mathbb{R})$, we have

$$\begin{aligned} [T]_{\mathcal{B}} &= \begin{bmatrix} [T(1)]_{\mathcal{B}} & [T(x)]_{\mathcal{B}} & [T(x^2)]_{\mathcal{B}} \end{bmatrix} \\ &= \begin{bmatrix} [1]_{\mathcal{B}} & [2x+1]_{\mathcal{B}} & [3x^2+2x]_{\mathcal{B}} \end{bmatrix} \\ &= \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{pmatrix}. \end{aligned}$$

Thus the characteristic polynomial of T is

$$\begin{aligned} \det([T]_{\mathcal{B}} - tI_3) &= \det \begin{pmatrix} 1-t & 1 & 0 \\ 0 & 2-t & 2 \\ 0 & 0 & 3-t \end{pmatrix} \\ &= (1-t)(2-t)(3-t) \\ &= (-1)(t-1)(t-2)(t-3) \end{aligned}$$

and hence $\lambda = 1, 2, 3$ are exactly the eigenvalues of T .

Theorem 5.3. : Let $A \in M_{n \times n}(\mathbb{F})$.

- (1) $f(t) := \det(A - tI_n)$ is a polynomial of degree n with leading coefficient $(-1)^n$.
- (2) A has at most n distinct eigenvalues.

Proof : (1)

Theorem 5.4. : Let $T : V \rightarrow V$ be a linear transformation of an n -dimensional vector space V .

Let λ be an eigenvalue of T and let $v \in V$ be a nonzero vector. Then

v is an eigenvector corresponding to λ if and only if $v \in N(T - \lambda I)$.

Example 4. : Let

$$A = \begin{pmatrix} 1 & 3 \\ 4 & 2 \end{pmatrix}.$$

$$(1) \det(A - tI_2) = \det \begin{pmatrix} 1-t & -3 \\ -4 & 2-t \end{pmatrix} = (1-t)(2-t) - 12 = t^2 - 3t - 10 = (t+2)(t-5)$$

Thus $t = -2$ and $t = 5$ are the eigenvalues of A .

$$(2) \text{ For } t = -2, \text{ let } (A + 2I_2) \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 & -3 \\ -4 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Then $x - y = 0$ and hence we can take $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

$$(3) \text{ For } t = 5, \text{ let } (A - 5I_2) \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -4 & -3 \\ -4 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Then $4x + 3y = 0$ and hence we can take $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}$.

Example 4. (Continued) : Let

$$A = \begin{pmatrix} 1 & 3 \\ 4 & 2 \end{pmatrix}.$$

(4) $v_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ and $v_2 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ are eigenvectors of A corresponding to $\lambda_1 = -2$ and $\lambda_2 = 5$.

(5) We have

$$\begin{pmatrix} 1 & 3 \\ 4 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = (-2) \times \begin{pmatrix} 1 \\ -1 \end{pmatrix} \text{ and } \begin{pmatrix} 1 & 3 \\ 4 & 2 \end{pmatrix} \begin{pmatrix} 3 \\ 4 \end{pmatrix} = 5 \times \begin{pmatrix} 3 \\ 4 \end{pmatrix}.$$

Thus

$$\begin{pmatrix} 1 & 3 \\ 4 & 2 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ -1 & 4 \end{pmatrix} = \left((-2) \times \begin{pmatrix} 1 \\ -1 \end{pmatrix} \ 5 \times \begin{pmatrix} 3 \\ 4 \end{pmatrix} \right) = \begin{pmatrix} 1 & 3 \\ -1 & 4 \end{pmatrix} \begin{pmatrix} -2 & 0 \\ 0 & 5 \end{pmatrix}.$$

In consequence, $A = \begin{pmatrix} 1 & 3 \\ 4 & 2 \end{pmatrix}$ can be written as

$$\begin{pmatrix} 1 & 3 \\ 4 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ -1 & 4 \end{pmatrix} \begin{pmatrix} -2 & 0 \\ 0 & 5 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ -1 & 4 \end{pmatrix}^{-1}$$

This shows that $A = \begin{pmatrix} 1 & 3 \\ 4 & 2 \end{pmatrix}$ is diagonalizable.

Example 6. : Find all the eigenvectors of

$$A = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix}.$$

$$(1) \ f(t) := \det \begin{pmatrix} 1-t & 1 \\ 4 & 1-t \end{pmatrix} = (1-t)^2 - 4 = t^2 - 2t - 3 = (t-3)(t+1)$$

Thus $\lambda = 3, -1$ are exactly the eigenvalues of A .

(2) Let $v = \begin{bmatrix} x \\ y \end{bmatrix}$ be an eigenvector of A corresponding to $\lambda = 3$.

$$\text{Then } \begin{pmatrix} -2 & 1 \\ 4 & -2 \end{pmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \text{ and so } -2x + y = 0. \text{ Thus } v = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ 2x \end{bmatrix} = x \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

In particular, $v = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ is an eigenvector of A corresponding to $\lambda = 3$.

(3) Let $v = \begin{bmatrix} x \\ y \end{bmatrix}$ be an eigenvector of A corresponding to $\lambda = -1$.

$$\text{Then } \begin{pmatrix} 2 & 1 \\ 4 & 2 \end{pmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \text{ and so } 2x + y = 0. \text{ Thus } v = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ -2x \end{bmatrix} = x \begin{bmatrix} 1 \\ -2 \end{bmatrix}.$$

In particular, $v = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$ is an eigenvector of A corresponding to $\lambda = -1$.

Example 6. (Continued) : Find all the eigenvectors of

$$A = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix}.$$

(4) Since $\left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \end{bmatrix} \right\}$ is a basis for $\mathbb{R}^2$, it follows that A is diagonalizable.

More precisely, we have

$$\begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = 3 \times \begin{pmatrix} 1 \\ 2 \end{pmatrix} \text{ and } \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -2 \end{pmatrix} = (-1) \times \begin{pmatrix} 1 \\ -2 \end{pmatrix}.$$

Thus

$$\begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 2 & -2 \end{pmatrix} = \left(3 \times \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad (-1) \times \begin{pmatrix} 1 \\ -2 \end{pmatrix} \right) = \begin{pmatrix} 1 & 1 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 0 & -1 \end{pmatrix}.$$

In consequence, $A = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix}$ can be written as

$$\begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 2 & -2 \end{pmatrix}^{-1}.$$

Remark : Let $T : V \rightarrow V$ be a linear transformation of an n -dimensional vector space V .

How to find the eigenvectors of $T : V \rightarrow V$?

Step 1. : Select an ordered basis $\mathcal{B} = \{v_1, \dots, v_n\}$ for V .

Then $A = [T]_{\mathcal{B}}$ is an $n \times n$ matrix.

Step 2. :

	V	$\xrightarrow{T}$	V	
	$\phi_{\mathcal{B}} \downarrow$		$\downarrow \phi_{\mathcal{B}}$	
	F^n	$\xrightarrow{L_A}$	F^n	.

Recall that $\phi_{\mathcal{B}}(v) = [v]_{\mathcal{B}}$ for $v \in V$.

Also it holds that $\phi_{\mathcal{B}} \circ T = L_A \circ \phi_{\mathcal{B}}$.

Step 3. : The characteristic polynomial of T is $f(t) := \det(A - tI_n)$.

Thus T and $A = [T]_{\mathcal{B}}$ have the same eigenvalues.

Let $\lambda \in \mathbb{F}$ be an eigenvalue of T and let $v \in V$. Then

v is an eigenvalue corresponding to λ

if and only if $\phi_{\mathcal{B}}(v)$ is an eigenvalue corresponding to λ .

Indeed, $T(v) = \lambda v \Leftrightarrow \phi_{\mathcal{B}} \circ T(v) = \phi_{\mathcal{B}}(\lambda v) \Leftrightarrow L_A(\phi_{\mathcal{B}}(v)) = \lambda \phi_{\mathcal{B}}(v)$.

In consequence, the problem of finding the eigenvectors of a linear operator $T : V \rightarrow V$

is reduced to the problem of finding the eigenvectors of a matrix.

Example 7. : Consider the linear operator

$$T : P_2(\mathbb{R}) \rightarrow P_2(\mathbb{R}), \quad T(f(x)) = f(x) + (x+1)f'(x),$$

For the standard basis $\mathcal{B} = \{1, x, x^2\}$ for $P_2(\mathbb{R})$, we have $[T]_{\mathcal{B}} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{pmatrix}$.

Also, $\lambda = 1, 2, 3$ are exactly the eigenvalues of $[T]_{\mathcal{B}}$.

For $\lambda_1 = 1$, let $B_1 = [T]_{\mathcal{B}} - I_3 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{pmatrix}$.

Then $v = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ is an eigenvector of $[T]_{\mathcal{B}}$ corresponding to $\lambda_1 = 1$ if and only if $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{pmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.

The latter gives us $y = 0$, $y + 2z = 0$ and $z = 0$.

Thus $v = \begin{bmatrix} a \\ 0 \\ 0 \end{bmatrix} = a \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ for some $a \in \mathbb{R} - \{0\}$.

Since $\phi_{\mathcal{B}}^{-1}(v) = a$, it follows that

the nonzero constant polynomials are the eigenvectors of T corresponding to $\lambda_1 = 1$.

Example 7. (Continued) :

For $\lambda_2 = 2$, let $B_2 = [T]_{\mathcal{B}} - 2I_3 = \begin{pmatrix} -1 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 1 \end{pmatrix}$.

Then $v = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ is an eigenvector of $[T]_{\mathcal{B}}$ corresponding to $\lambda_2 = 2$

$$\text{if and only if } \begin{pmatrix} -1 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 1 \end{pmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

$$\text{if and only if } v = \begin{bmatrix} a \\ a \\ 0 \end{bmatrix} = a \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \text{ for some } a \in \mathbb{R} - \{0\}.$$

Since $\phi_{\mathcal{B}}^{-1}(v) = a(x+1)$, it follows that

the eigenvectors of T corresponding to $\lambda_2 = 2$ are of the form $a(x+1)$ for some $a \in \mathbb{R} - \{0\}$.

Example 7. (Continued) :

For $\lambda_3 = 3$, let $B_3 = [T]_{\mathcal{B}} - 3I_3 = \begin{pmatrix} -2 & 1 & 0 \\ 0 & -1 & 2 \\ 0 & 0 & 0 \end{pmatrix}$.

Then $v = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ is an eigenvector of $[T]_{\mathcal{B}}$ corresponding to $\lambda_3 = 3$

$$\text{if and only if } \begin{pmatrix} -2 & 1 & 0 \\ 0 & -1 & 2 \\ 0 & 0 & 0 \end{pmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

$$\text{if and only if } v = \begin{bmatrix} a \\ 2a \\ a \end{bmatrix} = a \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \text{ for some } a \in \mathbb{R} - \{0\}.$$

Since $\phi_{\mathcal{B}}^{-1}(v) = a(x^2 + 2x + 1)$, it follows that the eigenvectors of T corresponding to $\lambda_3 = 3$

are of the form $a(x^2 + 2x + 1)$ for some $a \in \mathbb{R} - \{0\}$.

Example 7. (Continued) :

The eigenvectors $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$ form a basis for $\mathbb{R}^3$.

Thus we have

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

and hence

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}^{-1}.$$

Now, consider the ordered basis $\gamma := \{1, x+1, x^2+2x+1\}$ for $P_2(\mathbb{R})$.

Then we have

$$[T]_{\gamma} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

Remark : Let $T : V \rightarrow V$ be a linear transformation of an n -dimensional vector space V .

Let $\lambda \in \mathbb{F}$ be an eigenvalue of T and let $v \in V$ be an eigenvector of T corresponding to λ .

Let $W = \{av \mid a \in \mathbb{F}\}$ be the subspace of V spanned by v .

Then $T(av) = aT(v) = \lambda(av) \in W$ and hence we obtain the map $T : W \rightarrow W$.

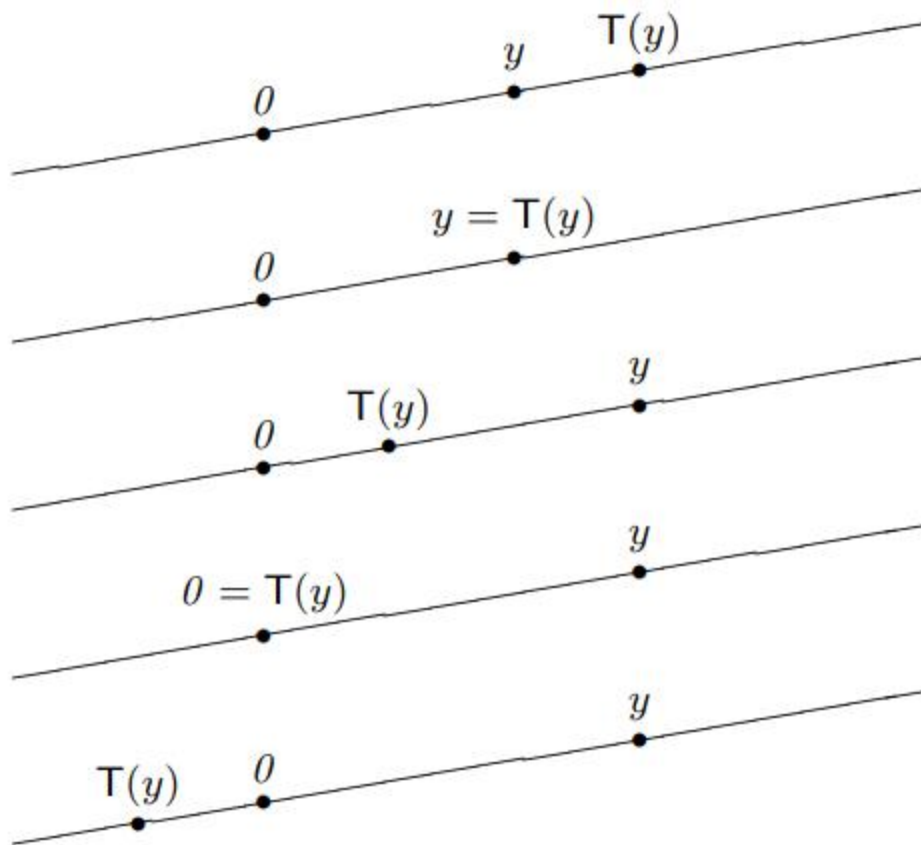
CASE 1. : $\lambda > 1$

CASE 2. : $\lambda = 1$

CASE 3. : $0 < \lambda < 1$

CASE 4. : $\lambda = 0$

CASE 5. : $\lambda < 0$



Example A. : The following linear operator is the reflection about the x -axis.

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^2, \quad (a_1, a_2) \mapsto (a_1, -a_2)$$

Thus $e_1 = (1, 0)$ and $e_2 = (0, 1)$ are eigenvectors of T .

Thus T acts as the identity on the x -axis and reverses the orientation of the y -axis.

Example B. : Suppose that $0 < \theta < \pi$, and

let $T_\theta : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the rotation of the plane $\mathbb{R}^2$ through the angle θ .

(1) $T_\theta : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ has no eigenvectors and therefore no eigenvalues.

(2) $T_\theta(a_1, a_2) = (a_1 \cos \theta - a_2 \sin \theta, a_1 \sin \theta + a_2 \cos \theta)$ and hence $[T]_{\{e_1, e_2\}} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$.

Then

$$\det([T]_{\{e_1, e_2\}} - tI_2) = \det \begin{pmatrix} \cos \theta - t & -\sin \theta \\ \sin \theta & \cos \theta - t \end{pmatrix} = t^2 - (2\cos \theta)t + 1$$

has no real zeros since $0 < \theta < \pi$.

Thus $T_\theta : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ has no eigenvalues and therefore no eigenvectors.